

QRM II Graded Assignment (9)

Period 1, 2026

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0 Introduction

This assignment is to be completed in groups of 4-5. Further, all students in your group need to be assigned to the same R tutorial group (Friday's tutorial). You can sign yourself up for a group on Canvas. Please do so **before the start of your first R tutorial on Friday September 4th**. You can use the Discussion Board in Canvas if you do not have a group yet or if your group is incomplete.

The assignment has 5 parts, and each part corresponds to the course material of that week (with the exclusion of week 6, for which there is no R programming material).

You are supposed to hand in these assignments on Canvas at the following dates:

- **Deadline 1** *Sunday September 27th, at 23:59pm*: you are supposed to hand in weeks 1, 2, and 3 of this assignment. This will determine 18% of your overall course grade
- **Deadline 2** *Sunday October 11th, at 23:59pm*: you are supposed to hand in weeks 4, and 5 of this assignment. This will determine 12% of your overall course grade

The R tutorials (each Friday) will consist of two halves. During the first half, you will discuss the tutorial exercises. These can be downloaded separately from Canvas. During the second half, you can work on this graded assignment within your own group. The purpose is that you find out how to work with R for doing statistical analyses by yourself. The tutorial exercises are meant to teach you basic commands to get you started, but to answer the problem sets in this assignment, you might need to research your own solutions, and use functions and commands not described in the tutorial exercises. Learning how to solve your own research problems is integral part of learning R. When you and your group get stuck on how to approach an exercise, the hierarchy in finding your way is as follows:

- use the concepts from the tutorial exercises;
- use the cheat sheets available on Canvas;
- use Google, YouTube, StackOverflow, or another website;
- ask the teacher.

The use of generative AI is **not** permitted and may result in a grade of 0. See the AI protocol in the course manual for details.

To answer the assignment, you can simply fill out this R markdown document. There are designated places which you can fill with R code. There are also designated spaces for you to answer each question. Often, the structure of an answer will be as follows. First, you type the R code in the designated box. This will show how you analyzed the data to get the answer to the question. Below the box for the R code, you will then summarize your answer to the question, i.e. what are the conclusions that you draw from the data analysis?

When handing in, you are supposed to submit this .Rmd file, and a knitted version of this document. You can knit this document to pdf, word, or html. Knitting to pdf requires you to have a .tex distribution installed on your computer. Knitting to Word requires you to have Word installed.

The exercises are designed such that you should be able to finish the majority of them during the tutorial each week. If you are not able to finish them fully during that time, you are expected to work on it in your own time using the computers on campus or your own device. It is best to meet as a group in-person when working together. If you want to work remotely, github is a good platform to guarantee smooth collaboration. Alternatively, you can email this .Rmd file back and forth to one another as a group, but this is not recommended as it is more cumbersome.

We encourage you to keep your code blocks, printing statements, and final answers, as short as possible. In any case, there is a page limit of 6 pages per week, which encompasses the total length of this document which consists of the questions, your coding lines, and your answers. When your answers to questions of the respective week exceed this page limit, they will not be graded, resulting in zero points.

1 Week 1

1. Find the dataset “movies3.tsv” on Canvas. Describe your data set: How many observations does it have. How many variables are there? How many subjects? What consists of a subject? **[4 points]**

#WRITE YOUR CODE HERE

Your Answer:

Write your response here.

2. Which of the following types of variables are present in your data set? (i) nominal; (ii) ordinal; (iii); continuous; (iv) discrete. If present, name one example of such a variable present in your data set. **[4 points]**

#WRITE YOUR CODE HERE

Your Answer:

Write your response here.

3. A movie studio wants to know which types of movies give maximal revenue. Perform the following steps to provide the movie studio with an analysis which corresponds to their request:
 - a. Summarize the variable revenue and report its mean, median, maximum, and minimum. **[2 points]**
 - b. Which movie has the highest revenue in your data set and how much is this revenue. Which movie has the lowest and how much is its revenue? If multiple movies have the exact same highest or lowest revenue, give only one example. **[2 points]**
 - c. Create a histogram of the variable revenue. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Describe the shape of the histogram. What does this tell you about the nature of making money in the movies industry? **[2 points]**
 - d. Add a new variable to your data set: the log of revenue. When creating this variable, what happens to movies for which revenue is zero? What then happens when you calculate the mean of log revenue? **[2 points]**
 - e. For movies that have a revenue of zero, replace log of revenue with “NA”. What is now the mean of log of revenue? Create a histogram for log revenue, again with an appropriate title, x- and y-axis labels. Describe the shape of this histogram. **[2 points]**

- f. For the variables revenue and log of revenue calculate their skewness and kurtosis. How do these numbers relate to the shape of the histograms that you estimated under d.) and e.)? **[2 points]**

For each step, you should provide first all the code you used to answer the question and then formulate an answer using full sentences.

Each week consists of 1, 2, or 3 subquestions. The total amount of points you can earn per week is 20 points.

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Step b

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Your Answer:

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2 Week 2

1 Is your dataset movies3.tsv the full population, or is it a sample of a larger population? If the latter, how would you describe the full population? [4 points]

Your answer here:

Write your formulated response here.

2 For this question, you will assume that your data set is the full population.

- What is the mean of the variable `vote_average` in your data set? [2 points]
- Create a new data set, called `movies_sample`. Make sure that it is a random sample of your data set of 25 movies. What is the mean of `vote_average` in this random sample? How does it compare to the mean of `vote_average` in 2a? [2 points]
- In a for loop, create 100 different samples of 25 movies, as in b, and estimate the mean within each sample. Save the mean of each sample in a vector called `sample_means`. So the first position of the vector would have the mean of the first sample, the second position the mean of the second sample, etc. Print the start of this vector. [2 points]
- Summarize and make a histogram of `sample_means`. What is the mean, standard deviation and shape of its distribution? [2 points]
- Using the formulas of the *Central Limit Theorem*, give the distribution of the sample mean of `vote_average` in a sample of 25 movies from your population (including its expected value and standard deviation). How do these values compare to your answers at d.)? [2 points]

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Your Answer:

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3. Is there a variable in your data set which you would expect to follow a discrete uniform distribution? Explain, without looking at the data, why you expect this data to be discrete uniform. If the data would indeed follow a discrete uniform distribution, what would you expect the mean and standard deviation of this variable to be? Next, plot a histogram of the variable, and plot an appropriate discrete uniform distribution on top of this histogram. Comment whether the variable indeed follows this expected distribution. If the distribution deviates from the discrete uniform distribution, explain why this is the case. **[6 points]**

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Your Answer:

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3 Week 3

For the next part of the assignment, assume that the movies in your data frame are a random sample of a larger population of movies.

1. Give a 90 percent confidence interval for the variable *revenue*. Interpret the result. **[2 points]**

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Your Answer:

Write your formulated response here.

2

- a. How many movies in your sample were directed by a male director? How many by a female director? **[2 points]**
- b. Test the null hypothesis that the chance that a movie is being directed by a male or female director is equal. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Use an appropriate R function to test your hypothesis, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**
- c. Recode revenue such that it is measured in millions of dollars. What is the variance of revenue for movies with a male director? What is the variance of revenue for movies with a female director? **[2 points]**

- d. Assume that the variance of revenue for movies with a male director in your data is the same as the variance for movies with a male director in your population (that is, there is no sampling uncertainty in this estimate). Now, set up a hypothesis test to test for the null hypothesis that the variance of revenue for movies with a female director is equal to the variance for movies with a male director. Clearly state your null hypothesis and alternative hypothesis, and your chosen significance level. Calculate your test statistic and tell how your test statistic is distributed, and state your conclusion. Also give the p-value that corresponds to your test statistic. **[2 points]**

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3

- a. Create a scatter plot with year on the x axis and mean of revenue within that year on the y-axis. Make sure it has an appropriate title, and appropriate titles and labels for the x- and y-axis. Is revenue of movies increasing or decreasing over time? **[3 points]**
- b. Recreate the scatter plot with year on the x axis and mean_revenue on the y-axis, but now add bars around each point, indicating the 95% confidence interval. **[3 points]**
- c. If you look closely at the resulting plot in b.), you will probably see that the confidence intervals in some years are wider than in other years. What are some possible reasons that a confidence interval in a given year can be wide? Can you verify this using your data? **[4 points]**

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4 Week 4

1. There is an argument going on in the movie studio. *Bob* claims that production budgets are getting out of hand, and that the studio should focus on making cheaper movies. *Chantal* disagrees. She tells Bob that “Every dollar we spend on movie production is more than offset by the increase in movie revenue”.

- a. Set up a regression model to test Chantal’s claim, and estimate it. That is, estimate:

$$\text{Revenue}_i = \beta_0 + \beta_1 \text{Budget}_i + \varepsilon_i.$$

Print a summary of your estimated model. **[2 points]**

- b. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- c. Test for the null hypothesis that $\beta_1 \leq 1$. Report the p-value and state your conclusion. **[2 points]**
- d. Next, estimate the model

$$\text{Log Revenue}_i = \beta_0 + \beta_1 \text{Log Budget}_i + \varepsilon_i.$$

When creating the variables Log Revenue and Log Budget, make sure that movies with a Revenue or Budget of zero are assigned the value “NA”. Print a summary of your estimated model. **[2 points]**

- e. What is the estimated value of β_1 and how do you interpret it? **[2 points]**
- f. Which model has better fit? The level-level model or the log-log model? Explain. **[2 points]**
- g. Who do you think is correct? Bob or Chantal? What would you advise the movie studio to do? **[2 points]**

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- a. Make a scatterplot with log revenue on the y-axis, and log budget on the x-axis. Within the same scatterplot, draw a straight line which summarizes the association between the log of revenue and log of budget. **[3 points]**
- b. Using the insights from the scatterplot, your linear model, and your data, find an example of a movie that vastly overperformed. That means, find a movie that had much more revenue than expected, given its budget. For this movie, report its log of revenue, its predicted log of revenue, and its residual. **[3 points]**

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5 Week 5

- Make a frequency table of the variable genre. Which are the three genres that occur the most? **[2 points]**
- Create a new variable called genre2. Make sure that the three most popular genres remain the same, but that all the other genres get categorized into one genre called “Other”. Report a frequency table of this new variable genre2. **[2 points]**
- Estimate an OLS model which has as dependent variable the log of revenue of a movie, and as independent variable the log of budget, a dummy for each level that genre2 can take, and the year of release. Show a summary of the resulting model and interpret each coefficient. **[4 points]**

The movie studio that you work at is releasing a new movie in 2026. It will be a Drama movie with a budget of 20,000,000. It will have a runtime of 120 minutes.

- Estimate a model that is able to predict the revenue of this movie. Give its predicted revenue and include a 90% prediction interval. **[5 points]**
- Plot the residuals against various independent variables in the model. Do you find any evidence of heteroskedasticity? **[2 points]**
- One executive at the studio wants to time the release of the movie to a specific month of the year such that they can maximize revenue. Another executive is of the opinion that the month in which the movie is released will not have a big impact on its revenue. Estimate a model that can test whether the month of release impacts the revenue of a movie. Use an appropriate hypothesis test to test whether month of release matters. What would you advise the movie studio regarding the timing of the release of the movie? **[5 points]**

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